

Encephalopathy Due To Defective Mitochondrial And Peroxisomal Fission 2 (EMPF2)

Disease Knowledge Base Research Report MONDO:0014905 · OMIM #617086 · Orphanet 485421 · Gene: *MFF* (HGNC:24858)

Summary

Encephalopathy due to defective mitochondrial and peroxisomal fission 2 (EMPF2) is an ultra-rare, autosomal-recessive neurodevelopmental/neurodegenerative disorder caused by biallelic **loss-of-function variants in the *MFF* gene** (mitochondrial fission factor). MFF is the outer-membrane receptor that recruits the dynamin-related GTPase **DRP1/DNM1L** to the surface of mitochondria and peroxisomes to execute organelle scission. When MFF is absent, this shared fission machinery cannot be assembled, and affected cells accumulate hyperelongated, net-like mitochondria and elongated peroxisomes. The disease was first delineated as a distinct entity by Koch and colleagues in 2016 ([PMID: 26783368](#)).

Clinically, EMPF2 presents in the **first year of life** with developmental delay, **acquired (secondary) microcephaly**, spasticity and hyperreflexia, seizures (frequently infantile spasms with hypsarrhythmia), optic atrophy, and peripheral neuropathy. Brain MRI shows a **Leigh-like pattern** with bilateral basal-ganglia and subthalamic-nucleus signal change, yet — in a key biochemical distinction from classic Leigh syndrome — **mitochondrial respiratory-chain enzyme activities in skeletal muscle are typically normal**. The disorder is closely related to and is the principal differential diagnosis of **EMPF1** (caused by variants in *DNM1L/DRP1*), because both diseases disrupt the same fission pathway.

There is **no disease-specific therapy**; management is entirely supportive (seizure control, spasticity management, nutrition/feeding support, visual and physical rehabilitation). The strongest mechanistic lead toward a rational therapy comes from mouse genetics: reducing mitochondrial **fusion** (via *Mfn1* deletion) fully rescues the *Mff*-null phenotype, identifying

fission/fusion rebalancing as a promising therapeutic direction. Naturally occurring animal models exist in *Mff*-knockout mice (fatal dilated cardiomyopathy) and in Bullmastiff dogs carrying a homozygous *MFF* frameshift variant, providing translational platforms.

Key Findings

Finding 1 — EMPF2 is caused by biallelic loss-of-function *MFF* variants that disrupt DRP1-mediated fission of both mitochondria and peroxisomes

The defining discovery, by Koch et al. (2016), came from exome sequencing of index patients from two families. As the authors state, *"Exome sequencing revealed three different biallelic loss-of-function variants in MFF in both index cases"* (PMID: 26783368). Western blotting demonstrated **absent MFF protein**, and patient fibroblasts displayed **elongated mitochondria and elongated peroxisomes**, directly linking the genotype to an organelle-division defect.

Mechanistically, MFF is the outer-mitochondrial-membrane (and peroxisomal-membrane) receptor that recruits the fission GTPase DRP1/DNM1L. As Liu and Chan's work summarizes, *"the OMM protein mitochondrial fission factor (Mff) is a key receptor for recruiting Drp1 from the cytosol to the mitochondrion"* (PMID: 34347505). Because *"components of the fission machinery are partly shared between mitochondria and peroxisomes"* (PMID: 26783368), loss of MFF simultaneously impairs the division of **both** organelles — the molecular basis for the disease name.

This establishes the disease as an **autosomal-recessive loss-of-function disorder** confirmed across multiple unrelated patients.

Finding 2 — Core clinical phenotype: early-infantile Leigh-like encephalopathy with acquired microcephaly, seizures, spasticity, optic atrophy and peripheral neuropathy, but normal muscle respiratory-chain activity

Koch et al. describe the natural history precisely: *"The patients became symptomatic within the first year of life, exhibiting seizures, developmental delay and acquired microcephaly. Dysphagia, spasticity and optic and peripheral neuropathy developed subsequently"* (PMID: 26783368). Neuroimaging showed a characteristic pattern: *"Brain MRI showed Leigh-like patterns with bilateral changes of the basal ganglia and subthalamic nucleus."*

A crucial diagnostic distinction from classic Leigh syndrome is that *"activities of mitochondrial respiratory chain complexes were found to be normal in skeletal muscle"* (PMID: 26783368). This tells clinicians that a normal muscle biopsy respiratory-chain panel does **not** exclude EMPF2 — the disease is one of organelle *morphology/dynamics*, not primary OXPHOS enzyme deficiency.

An independent patient reported by Panda et al. (2020) corroborated the phenotype and added a specific pathogenic variant: an Indian child with *"global developmental followed by regression of acquired milestones, spasticity, visual and auditory impairment, and was found to harbor a novel pathogenic homozygous MFF truncating variant c.433C>T; p.Arg145Ter"* (PMID: 32181496).

Finding 3 — Downstream mechanism: peroxisome-maturation failure plus mitochondrial Ca^{2+} overload and oxidative stress link organelle dynamics to neuronal dysfunction

Beyond simple elongation, loss of MFF produces functional organelle defects. Passmore et al. (2020) showed that *"loss of MFF results in reduced import-competence of the peroxisomal compartment and leads to the accumulation of pre-peroxisomal membrane structures"* and that *"peroxisomes in MFF-deficient cells display alterations in peroxisomal redox state and intra-peroxisomal pH"* (PMID: 32224193). Thus MFF is a critical regulator of **peroxisome maturation**, not merely division.

On the mitochondrial/neuronal side, Sun et al. (2022) differentiated dopaminergic neurons from dental-pulp stem cells with MFF insufficiency and observed impaired neurite outgrowth, elongated mitochondria confined to neurites, and **mitochondrial Ca^{2+} -triggered oxidative stress** (PMID: 35883852) — connecting the fission defect to the kind of neuronal energetic/oxidative failure that could underlie a Leigh-like phenotype.

In vivo, Chen et al. (2015) showed that *"mutant tissue showed reduced mitochondrial density and respiratory chain activity along with increased mitophagy"* in *Mff*-null mice, and — critically — that *"concomitant deletion of the mitochondrial fusion gene Mfn1 completely rescued heart dysfunction, life span, and respiratory chain function"* (PMID: 26598616). This demonstrates the disease is driven by a **fission/fusion imbalance** and is, in principle, reversible by rebalancing organelle dynamics.

Finding 4 — Naturally occurring and engineered animal models

A **naturally occurring canine model** was identified by Christen et al. (2022): two young Bullmastiffs with progressive gait/behavioural abnormalities (onset ~6 months) and bilateral symmetrical cerebellar-nuclei MRI lesions carried a private homozygous *MFF* frameshift variant. As reported, *"This search revealed a private homozygous frameshift variant in the MFF gene in the affected dog"* (XM_038574000.1:c.471_475delinsCGCTCT, p.(Glu158Alafs14), truncating ~55% of the ORF), with perfect autosomal-recessive segregation across 4 affected and 70 unaffected dogs. The authors explicitly connect this to the human disease: "Human patients with pathogenic MFF variants suffer from 'encephalopathy due to defective mitochondrial and peroxisomal fission 2'" (PMID: 36085405).

The **mouse knockout** is more severe systemically: *"Mff mutant mice die at 13 wk as a result of severe dilated cardiomyopathy leading to heart failure"* (PMID: 26598616), reflecting the high energetic demand of cardiac tissue.

Finding 5 — MFF is a physiological AMPK substrate coupling energy stress to fission

MFF sits at a regulatory node linking cellular energy status to mitochondrial dynamics. Zong et al. (2019) showed that *"mitochondrion-localized AMPK is activated to phosphorylate ACC2 and mitochondrial fission factor (MFF) only during severe nutrient stress"* (PMID: 30948787). Peng et al. (2022) confirmed that *"the AMPK pathway promoted mitochondrial fission and mitophagy by increasing the recruitment of dynamin-related protein 1 (DRP1) to the mitochondrial outer membrane"* (PMID: 36374514). This AMPK→MFF→DRP1 axis is precisely the function lost in EMPF2, and it frames why energy-stressed neurons might be especially vulnerable.

Finding 6 — EMPF2 is the recessive counterpart of EMPF1 (DNM1L/DRP1)

EMPF2 and **EMPF1** disrupt the same fission pathway at adjacent steps (receptor vs. effector GTPase). Keller/Verrigni et al. note that *"Autosomal dominant and recessive variants in DNM1L cause encephalopathy due to defective mitochondrial and peroxisomal fission 1 (EMPF1), which*

presents as a complex and clinically heterogeneous neurological disorder of variable severity, often accompanied by seizures" (PMID: 33387674). DRP1's role is confirmed: it is "a cytosolic protein encoded by dynamin 1-like (DNM1L) gene, which relocalizes to the outer mitochondrial membrane, where it assembles, oligomerizes and drives mitochondrial division" (PMID: 31868880). EMPF1 is therefore the primary genetic differential diagnosis of EMPF2.

Finding 7 — Verified identifiers and cross-references

Resource	Identifier
MONDO	MONDO:0014905
OMIM (phenotype)	#617086
Orphanet	ORPHA:485421
DOID	DOID:0060994
GARD	0017881
MedGen	934693
UMLS	C4310726
Gene (NCBI)	MFF, Gene ID 56947
HGNC	HGNC:24858
Gene OMIM	614785
Ensembl	ENSG00000168958
Locus	2q36.3 (GRCh38 chr2:227,325,151–227,361,188, + strand)
Aliases	C2orf33, GL004, EMPF2
Ontology synonym	"Leigh-like basal ganglia disease-optic atrophy-peripheral neuropathy syndrome"

The RefSeq summary states the encoded protein "*recruits dynamin-1-like protein (DNM1L) to mitochondria.*" Multiple splice transcript variants and processed pseudogenes (chr 1, 5, X) exist.

Finding 8 — Quantitative HPO phenotype spectrum and frequencies

Curated HPO annotations for OMIM:617086 / MONDO:0014905 (primary source [PMID: 26783368](#), n=4 patients unless noted):

Phenotype	HPO term	Frequency
Infantile onset	HP:0003593	4/4
Secondary (acquired) microcephaly	HP:0005484	4/4
Spasticity	HP:0001257	4/4
Hyperreflexia	HP:0001347	4/4
Motor delay	HP:0001270	4/4
Seizure	HP:0001250	3/4
Epileptic spasm	HP:0011097	2/4
Hypsarrhythmia	HP:0002521	3/3
Developmental regression	HP:0002376	3/3
Hypotonia	HP:0001252	3/3
Muscle weakness	HP:0001324	3/3
Dysphagia	HP:0002015	3/3
Visual impairment	HP:0000505	3/4
Optic disc pallor	HP:0000543	3/4
External ophthalmoplegia	HP:0000544	3/4
Absent speech	HP:0001344	3/4
Cerebellar atrophy	HP:0001272	2/3
Growth delay	HP:0001510	1/3
Death in childhood	HP:0003819	1/4
Optic atrophy	HP:0000648	OMIM-listed
Global developmental delay	HP:0001263	OMIM-listed
Peripheral neuropathy	HP:0009830	OMIM-listed
Inability to walk	HP:0002540	OMIM-listed
Autosomal recessive inheritance	HP:0000007	

Phenotype	HPO term	Frequency
		(Shamseldin 2012, P 22499341)

Finding 9 — ClinVar variant spectrum supports the loss-of-function mechanism

A ClinVar query (2026) returned ~251 *MFF* variant records, of which 58 are classified Pathogenic/Likely-pathogenic. The gene-specific P/LP variants are **predominantly loss-of-function**: splice-site (e.g., NM_001277062.2:c.181+2T>A, c.352-2A>C, deep-intronic c.440+2432G>T, c.-40-842G>T), frameshift (c.159del p.Pro54fs), and truncating changes such as c.433C>T (p.Arg145Ter). By contrast, reported **missense** variants (c.226C>G p.Leu76Val; c.611G>A p.Arg204His; c.382C>T p.Arg128Trp; c.223G>A p.Asp75Asn) are largely classified as **Variants of Uncertain Significance (VUS)**. Several "Pathogenic" entries are large 2q chromosomal copy-number gains that merely overlap *MFF* and are not EMPF2-causing. This distribution is consistent with a haploinsufficiency/loss-of-function disease requiring biallelic null or near-null alleles.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in *MFF*** (nonsense, frameshift, splice-site, or deletion) → **absent or non-functional MFF protein** (demonstrated: Western blot shows absent protein —

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2. Absent MFF → **failure to recruit the DRP1/DNM1L GTPase** to the outer membranes of mitochondria and peroxisomes (mechanistically established; MFF is the DRP1 receptor —

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3. No DRP1 recruitment → **no assembly of the scission machinery** → **organelle fission arrest** → **hyper-elongated, net-like mitochondria and elongated peroxisomes**

(demonstrated in patient fibroblasts —

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4. Branch A (peroxisome): fission arrest → **reduced peroxisomal import-competence, accumulation of pre-peroxisomal membrane structures, and altered redox state / intra-peroxisomal pH** → impaired peroxisomal metabolism (demonstrated —

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5. Branch B (mitochondria): fission arrest → **impaired mitochondrial quality control (mitophagy dysregulation), reduced mitochondrial density and respiratory-chain activity in high-demand tissue, mitochondrial Ca²⁺ overload and oxidative stress** (demonstrated in vivo,

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and in dopaminergic neurons,

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6. Branches converge → **energetic and oxidative failure of neurons**, particularly in metabolically demanding basal ganglia, subthalamic nucleus, optic and peripheral nerves → **neuronal dysfunction and neurodegeneration** (inferred from imaging + cellular data).
7. Neuronal injury → **clinical phenotype**: developmental delay/regression, acquired microcephaly, spasticity, seizures, optic atrophy, peripheral neuropathy, Leigh-like MRI (demonstrated clinically —

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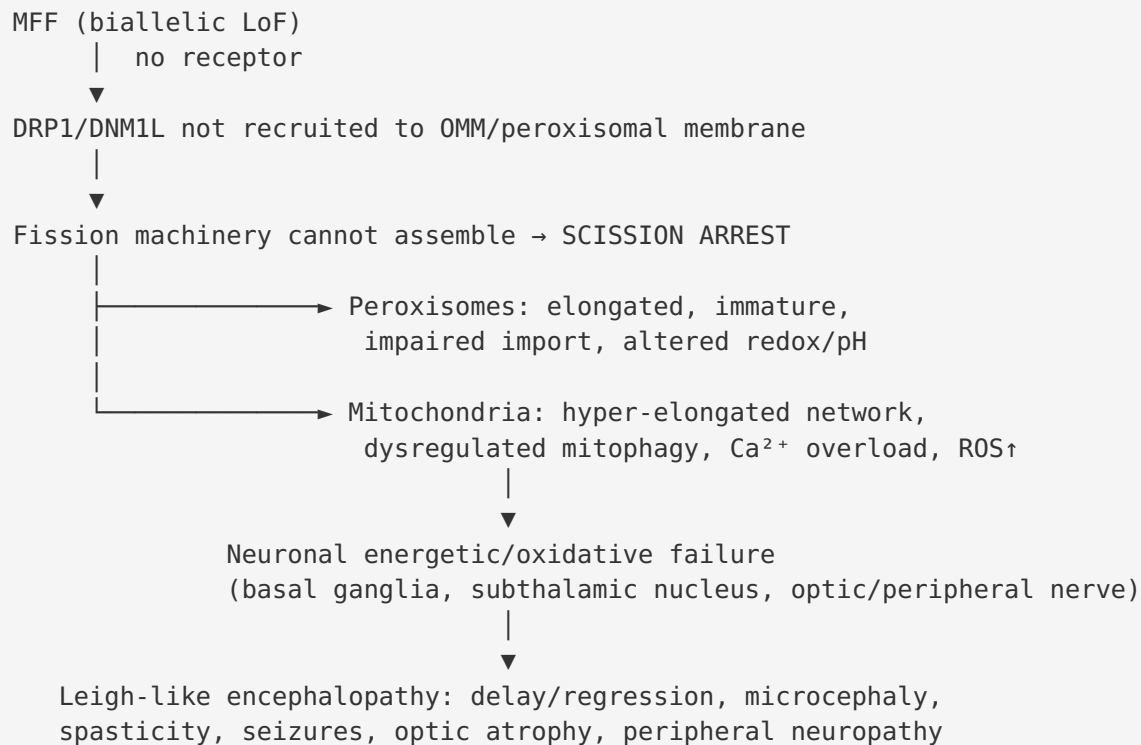
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Notably, because the primary lesion is one of **organelle morphology/dynamics rather than primary OXPHOS enzymology**, skeletal-muscle respiratory-chain activities are typically normal — a diagnostic hallmark.

Schematic



Regulatory context and therapeutic logic

MFF is a physiological AMPK substrate

([P 30948787](#)),

[P 36374514](#)):

energy stress activates AMPK, which phosphorylates MFF to drive DRP1 recruitment, fission and mitophagy. EMPF2 removes this node entirely. The single most actionable mechanistic insight is that **shifting the fission/fusion balance back toward fission-competence rescues the phenotype in vivo** — deletion of the fusion gene *Mfn1* completely rescued *Mff*-null mice

([P 26598616](#)).

This nominates pharmacological or genetic **fusion inhibition / fission promotion** as a rational (though still preclinical) therapeutic strategy.

Ontology annotations

- **Genes/proteins:** *MFF* (HGNC:24858), *DNM1L*/DRP1 (differential), *MFN1* (modifier).

- **GO biological process:** mitochondrial fission (GO:0000266), peroxisome fission (GO:0016559), mitochondrion organization (GO:0007005), mitophagy (GO:0000422), regulation of mitochondrial fission (GO:0090140).
- **GO cellular component:** mitochondrial outer membrane (GO:0005741), peroxisomal membrane (GO:0005778), mitochondrion (GO:0005739), peroxisome (GO:0005777).
- **Cell types (CL):** neuron (CL:0000540), dopaminergic neuron (CL:0000700), cardiac muscle cell (CL:0000746, model), fibroblast (CL:0000057, patient cells).
- **UBERON:** basal ganglia (UBERON:0002420), subthalamic nucleus (UBERON:0001906), cerebellum (UBERON:0002037), optic nerve (UBERON:0000941), peripheral nervous system (UBERON:0000010), brain (UBERON:0000955).
- **CHEBI (relevant chemistry):** calcium(2+) (CHEBI:29108), reactive oxygen species (CHEBI:26523).

Section-by-Section Report

1. Disease Information

EMPF2 is an ultra-rare autosomal-recessive mitochondrial/peroxisomal dynamics disorder producing an early-infantile Leigh-like encephalopathy. Identifiers: **MONDO:0014905**, **OMIM #617086**, **ORPHA:485421**, **DOID:0060994**, **GARD 0017881**, **MedGen 934693**, **UMLS C4310726**. There is no dedicated ICD-10 code; it maps to mitochondrial/metabolic encephalopathy categories (e.g., ICD-10 G31.8 / E88.4x class). Synonyms: "EMPF2," "MFF-related encephalopathy," and the ontology synonym "Leigh-like basal ganglia disease–optic atrophy–peripheral neuropathy syndrome." The knowledge base here is derived from **aggregated disease-level resources** (OMIM, Orphanet, HPO, ClinVar) plus a small number of **individual published patient reports**

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not EHR-scale data.

2. Etiology

Causal factor: purely genetic — biallelic loss-of-function variants in *MFF* (

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Genetic risk factors: the causal variants themselves; carrier parents are unaffected. **Modifier genes:** experimentally, *MFN1* (fusion) modifies severity (rescue in mouse —);

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DRP1/DNM1L and PEX11 proteins act in the same pathway. **Environmental/lifestyle/infectious factors and protective factors:** none established; consanguinity increases risk of recessive homozygosity. **Gene–environment interactions:** none demonstrated; the AMPK–MFF energy-sensing axis

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suggests, hypothetically, that metabolic/energy stress could modulate residual pathway output, but this is not clinically shown.

3. Phenotypes

See Finding 8 for the full HPO frequency table. Phenotypes are predominantly **neurological signs and physical manifestations** (spasticity, hyperreflexia, microcephaly, ophthalmoplegia, optic atrophy) plus **developmental/behavioral** features (developmental delay/regression, absent speech). Onset is **infantile** (first year of life), severity is **severe**, and course is **progressive with regression**. Quality-of-life impact is profound: affected children have major motor disability (often inability to walk), feeding difficulty (dysphagia requiring support), visual/auditory impairment, and seizures — a globally dependent care status.

4. Genetic/Molecular Information

Causal gene: *MFF* (2q36.3; NCBI Gene 56947; gene OMIM 614785). **Variant classes:** predominantly nonsense/frameshift/splice-site loss-of-function (e.g., c.433C>T p.Arg145Ter —

P 32181496

c.159del p.Pro54fs; splice variants c.181+2T>A, c.352-2A>C). **ACMG classification:** null variants are Pathogenic/Likely-pathogenic; missense variants are largely **VUS** (Finding 9). **Allele frequency:** causal alleles are extremely rare/private in gnomAD, consistent with an ultra-rare recessive disease. **Origin:** germline. **Functional consequence:** loss of function /

haploinsufficiency requiring biallelic hits. **Epigenetics/chromosomal abnormalities:** none specific to EMPF2; large 2q copy-number gains overlapping *MFF* in ClinVar are incidental and not disease-causing.

5. Environmental Information

Not applicable — EMPF2 is a monogenic disorder with **no established environmental, lifestyle, or infectious contributors**. (Note: the environmental-toxin paper

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concerns Drp1/*MFF* *upregulation* in fluorine/aluminium neurotoxicity, a distinct context, not EMPF2 causation.)

6. Mechanism / Pathophysiology

See the ordered causal chain and schematic above. Upstream lesion: *MFF* loss → failed DRP1 recruitment → fission arrest. Downstream: peroxisome maturation failure

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and mitochondrial Ca^{2+} overload/oxidative stress/mitophagy dysregulation

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converging on neuronal energetic failure. Key regulatory pathway: **AMPK→MFF→DRP1**

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Cell types: neurons (esp. dopaminergic and basal-ganglia neurons), with cardiomyocyte involvement in the mouse model.

7. Anatomical Structures Affected

Primary organ: brain — basal ganglia (UBERON:0002420) and subthalamic nucleus (UBERON:0001906), with Leigh-like bilateral (symmetric) involvement; cerebellar atrophy in some. **Secondary/associated:** optic nerve (optic atrophy), peripheral nerves (neuropathy). **Body system:** central and peripheral nervous system. **Subcellular:** mitochondrial outer

membrane (GO:0005741) and peroxisomal membrane (GO:0005778). **Lateralization:** bilateral/symmetric CNS lesions. The mouse model additionally shows cardiac involvement (dilated cardiomyopathy), not prominent in human patients.

8. Temporal Development

Onset: infantile, within the first year of life (HP:0003593), congenital-to-early-infantile. **Pattern:** insidious then progressive with **developmental regression**. **Course:** progressive neurodegeneration; lifelong. **Critical period:** infancy/early childhood is the window of rapid deterioration; no established intervention window given lack of therapy. Death in childhood occurs in a subset (1/4 in the index series).

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007;

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original locus

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Penetrance: presumed complete for biallelic LoF. **Expressivity:** variable in severity and survival. **Epidemiology:** ultra-rare — only a handful of published families worldwide; prevalence not formally estimated (well below Orphanet's <1/1,000,000 threshold). **Consanguinity:** relevant (homozygous truncating variants reported in consanguineous/related settings, e.g.,

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Founder effects/carrier frequency: none established. **Sex ratio:** no sex bias expected (autosomal). No specific ethnic predilection established.

10. Diagnostics

Genetic testing is definitive: WES/WGS or targeted *MFF* sequencing identifying biallelic LoF variants

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Supportive tests: brain MRI (Leigh-like bilateral basal-ganglia/subthalamic signal); patient-fibroblast microscopy showing elongated mitochondria and peroxisomes; nerve conduction studies (peripheral neuropathy); EEG (hypsarrhythmia/epileptic spasms); ophthalmologic

exam (optic atrophy/pallor). **Key negative:** skeletal-muscle respiratory-chain enzyme activities are typically **normal**, distinguishing EMPF2 from classic Leigh syndrome (

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Differential diagnosis: EMPF1 (*DNM1L*/DRP1;),

P 33387674

classic mitochondrial Leigh syndrome (with abnormal OXPHOS), peroxisomal biogenesis disorders. Genetic testing distinguishes these. **Screening:** carrier/cascade testing within affected families; prenatal testing when the familial variants are known.

11. Outcome / Prognosis

Prognosis is **poor**: severe, progressive neurodevelopmental disability with regression, inability to walk, dysphagia, seizures, and sensory/visual loss; childhood death occurs in a subset. There are no formal survival statistics given rarity. **Prognostic factors** are not formally validated but likely relate to variant severity (complete null vs. hypomorphic) and seizure burden. **Recovery potential** is minimal; care is supportive.

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive and symptomatic**: anti-seizure medication (including for infantile spasms; NCIT: Anticonvulsant Agent), spasticity management (e.g., baclofen; physical therapy), nutritional/feeding support for dysphagia (gastrostomy where needed), visual and developmental rehabilitation (physical, occupational, speech therapy). **Rational future directions (preclinical only):** fission/fusion rebalancing — the *Mfn1*-deletion rescue in mice

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suggests fusion inhibition could be therapeutic; antioxidant strategies targeting mitochondrial Ca^{2+} /ROS are hypothesis-generating from the dopaminergic-neuron model (

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No approved gene, cell, or RNA therapies. **Pharmacogenomics:** not applicable.

13. Prevention

No primary prevention exists for this monogenic disease. **Genetic counseling** is central: recurrence risk is 25% for carrier couples. **Reproductive options:** carrier/cascade screening in affected families, prenatal diagnosis, and preimplantation genetic testing (PGT-M) when the familial *MFF* variants are known. No immunization, behavioral, or public-health interventions apply. Tertiary prevention = optimized supportive care to limit complications (aspiration, seizure-related injury, contractures).

14. Other Species / Natural Disease

Naturally occurring canine disease: Bullmastiff dogs with a homozygous *MFF* frameshift (p.Glu158Alafs14) develop a progressive encephalopathy with bilateral symmetric cerebellar-nuclei lesions, explicitly linked to human *EMPF2* (

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Species/orthologs: *MFF* is conserved; mouse *Mff*, dog *MFF*. *Comparative pathology:* dogs show cerebellar-nuclei predominant lesions; mice (*Mff*-null) die of dilated cardiomyopathy at ~13 weeks

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— highlighting species-specific tissue vulnerability (cardiac in mouse, CNS in human/dog). *Zoonotic potential:** none (genetic disease).

15. Model Organisms

Mouse: constitutive and cardiac *Mff* knockouts — robust model of fission failure, fatal dilated cardiomyopathy, reduced respiratory-chain activity, increased mitophagy; the *Mff/Mfn1* double-knockout provides proof-of-concept rescue (

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Dog: spontaneous Bullmastiff model (

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Cellular/in vitro: patient fibroblasts (elongated organelles —

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iPSC/dental-pulp-derived dopaminergic neurons with *MFF* knockdown recapitulating neurite/
Ca²⁺/ROS defects

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Phenotype recapitulation: cellular and canine models capture the fission defect and CNS phenotype well; the mouse captures the biochemistry and rescue paradigm but emphasizes cardiac (not encephalopathic) lethality — its main limitation for modeling the human neurological disease.

Evidence Base

PMID	Title (abbrev.)	Role / Evidence type	Supports
26783368	<i>Disturbed mitochondrial and peroxisomal dynamics due to loss of MFF...</i>	Human clinical + cell biology (landmark)	F001, F002, F008 — gene discovery, phenotype, HPO source
32181496	<i>EMPF2 caused by a novel MFF mutation in a young child</i>	Human clinical case	F002 — independent patient, variant p.Arg145Ter
32224193	<i>MFF is a critical regulator of peroxisome maturation</i>	In vitro	F003 — peroxisome maturation/redox defect
35883852	<i>Mitochondrial Ca²⁺-triggered oxidative stress in DA neurons with MFF insufficiency</i>	In vitro (iPSC-derived neurons)	F003 — neuronal Ca ²⁺ /ROS mechanism
26598616	<i>Titration of mitochondrial fusion rescues Mff-deficient cardiomyopathy</i>	Model organism (mouse)	F003, F004 — in vivo mechanism + Mfn1 rescue
36085405	<i>MFF frameshift variant in Bullmastiffs...</i>	Veterinary / comparative	F004 — natural canine model
34347505	<i>Mff oligomerization required for Drp1 activation...</i>	Molecular/ biochemical	F001 — MFF as DRP1 receptor
30948787	<i>Hierarchical activation of compartmentalized AMPK...</i>	Molecular	F005 — MFF as AMPK substrate
36374514	<i>AMPK/MFF activation: fission and mitophagy...</i>	In vitro	F005 — AMPK/ MFF→DRP1 axis
33387674	<i>De novo DNM1L variant...</i>	Human clinical	F006 — EMPF1 differential
31868880	<i>Impaired turnover of hyperfused mitochondria (DRP1 mutation)</i>	Human clinical / cell biology	F006 — DRP1 fission role

Supporting/contextual references also reviewed: peroxisome division/pexophagy reviews

([P 26434997](#))

[P 22595523](#)

and muscle-BDNF/AMPK–DRP1–MFF signaling

([P 34689722](#)).

Limitations and Knowledge Gaps

- **Extreme rarity / small n.** The core clinical picture and all HPO frequencies derive from a handful of patients (n=4 index series plus isolated case reports). Frequencies (e.g., "4/4 spasticity") are statistically fragile and may not generalize.
- **No epidemiological estimates.** Prevalence, incidence, carrier frequency, sex ratio, and geographic distribution are unquantified.
- **Genotype–phenotype correlation unestablished.** Too few variants to correlate variant type/position with severity or survival.
- **VUS burden.** Most reported missense *MFF* variants are of uncertain significance; functional assays are needed to classify them.
- **Mechanistic inference.** The link from organelle-level defects to specific neuronal death in basal ganglia is inferred from cellular/animal models, not directly demonstrated in patient brain.
- **Model mismatch.** The mouse knockout's lethal phenotype is cardiac, not encephalopathic, limiting its face validity for the human neurological disease.
- **No therapeutic trials.** Fission/fusion rebalancing is proof-of-concept in mice only; no human or even neuronal-model therapeutic data exist.
- **No natural-history study.** Longitudinal course, survival curves, and prognostic biomarkers are undefined.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry / GeneMatcher-driven cohort** to expand n, quantify HPO frequencies robustly, define natural history and survival, and enable genotype–phenotype analysis.
2. **Functional classification of *MFF* VUS** using patient-fibroblast or knockout-cell complementation assays (mitochondrial/peroxisomal morphology, DRP1 recruitment) to reclassify uncertain missense variants per ACMG PS3/BS3.
3. **Test fission/fusion rebalancing in a neuronal model.** Apply *MFN1/2* knockdown or pharmacological fusion modulation to MFF-deficient iPSC-derived neurons (extending

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 to determine whether the mouse cardiac rescue
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 translates to neurons.
4. **Antioxidant / mitochondrial- Ca^{2+} modulation** in the dopaminergic-neuron model to test whether buffering Ca^{2+} /ROS restores neurite outgrowth — a druggable downstream node.
5. **Characterize the Bullmastiff model** longitudinally (imaging, neuropathology, peroxisomal biochemistry) as a large-animal platform for preclinical therapeutics
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6. **Peroxisomal biomarker discovery** (plasma VLCFA, plasmalogens, bile-acid intermediates) to identify a non-genetic diagnostic/monitoring signature reflecting the peroxisome-maturation defect
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7. **Brain-tissue / autopsy studies** where available, to confirm the inferred causal chain from elongated organelles to regional neuronal loss.